Algorithmic Long-only Strategy Development on Indian Equity Market Data

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Documentation

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1. Previous Approaches: Sector Classification

1.1 Approach 1 — Dual-Index Construction (Market-Cap & Inverse-Volatility)

The Approach: Rather than a single sector proxy, we constructed **two complementary internal indices** per sector to capture both market-cap dominance *and* risk-adjusted breadth.

Index A: Market-Cap Weighted (MCW)

Let $M_i(t)$ be the market capitalisation of stock i at time t, and let $M_0 = \sum_i M_i(t_0)$ be the aggregate cap at the base date with index value $I_0 = 100$. Then:

$$I_{\text{MCW}}(t) \;=\; 100 \times \frac{\sum_i M_i(t)}{M_0}$$

Large-caps dominate this index (historically $\sim$40% of movement comes from the top 5 names), making it necessary to supplement with a risk-adjusted lens.

Index B: Inverse-Volatility Weighted (IVW)

For each stock compute the annualised volatility from daily log-returns $r_i^{(\tau)} = \ln\!\big(P_i^{(\tau)}/P_i^{(\tau-1)}\big)$:

$$\sigma_i^{\text{ann}} = \sqrt{252} \;\cdot\; \text{std}\big(r_i^{(\tau)}\big)$$

Raw inverse-vol weights, normalised to sum to unity:

$$w_i = \frac{1/\sigma_i^{\text{ann}}}{\sum_j 1/\sigma_j^{\text{ann}}}$$

The index compounds daily using the weighted log-return:

$$I_{\text{IVW}}(t) = I_{\text{IVW}}(t-1) \cdot \exp\!\Big(\sum_i w_i\, r_i(t)\Big)$$

Why We Pivoted: Whilst the dual-index correctly identified sector-level health, it gave no mechanism for *intra-sector* stock selection. Two sectors with identical dual-index readings could contain vastly different stock distributions. We also found that sudden volatility spikes temporarily collapsed w_i, destabilising the IVW signal for 5–10 days post-shock.

1.2 Approach 2 — Z-Score Cross-Sectional Ranking

The Approach: Using the dual-index idea, we computed Z-scores for both MCW returns $\hat{R}_i^{\text{MC}}$ and IVW ("Smart Beta") returns $\hat{R}_i^{\text{SB}}$ at the *individual-stock* level, then formed a composite signal:

$$Z_i^{\text{MC}} = \frac{\hat{R}_i^{\text{MC}} - \mu^{\text{MC}}}{\sigma^{\text{MC}}}, \qquad Z_i^{\text{SB}} = \frac{\hat{R}_i^{\text{SB}} - \mu^{\text{SB}}}{\sigma^{\text{SB}}}, \qquad \text{Score}_i = \alpha\, Z_i^{\text{MC}} + (1-\alpha)\, Z_i^{\text{SB}}, \quad \alpha \in [0,1]$$

Why We Pivoted: Z-scores assume Gaussian returns. Indian equities exhibit pronounced fat tails ($\kappa > 3$). A single outlier day inflated Z_i^{MC} dramatically, generating false-positive buy signals immediately before mean-reversion. Tuning α across regimes required look-ahead, violating out-of-sample integrity.

2. Previous Approaches: Top-Down Sector Allocation

2.1 Approach 3 — Adapted Ranked Asset Allocation Model (RAAM)

The Approach: We treated all 11 GICS sectors as distinct asset classes and ranked them monthly using a composite score built on three sub-signals:

Signal	Definition	Lookback
Absolute Momentum	$\mu_s = R_s^{4\text{m}} - r_f$ (go to cash if $\mu_s < 0$)	4 months
GARCH Volatility	$\hat{\sigma}_s^{\text{GARCH}}$ smoothed over 10-day EMA	10 days
Relative Correlation	$\bar{\rho}_s = \frac{1}{S-1} \sum_{s' \neq s} \rho_{s,s'}$	rolling 60d

The composite RAAM score and capital rule:

$$\text{RAAM}_s = \omega_1 \, \mu_s - \omega_2 \, \hat{\sigma}_s^{\text{GARCH}} - \omega_3 \, \bar{\rho}_s, \qquad \text{allocate to } \mathcal{S}^* = \arg\operatorname*{top5}_s \text{ RAAM}_s$$

An ATR-based breakout filter timed entries within each selected sector, and a binary risk-off rule forced a sector entirely to cash when $\mu_s < 0$.

Why We Pivoted: Two fundamental failures emerged:

1. **Intra-Sector Dispersion Trap.** Forcing capital into every stock within a "Top-5" sector purchased weak, lagging names alongside leaders. The sector label masked massive intra-sector dispersion, producing drag that erased the inter-sector alpha edge.
2. **Equity Correlation Illusion.** RAAM is designed for *cross-asset* diversification where $\rho_{s,s'} \ll 1$. For purely equity-based sectors, systemic market risk drives $\rho_{s,s'} \to 1$ during drawdowns—precisely when diversification is most needed. The low-correlation ranking signal became uninformative under stress.

$\rightarrow$ **Solution:** These twin failures forced a pivot from top-down sector allocation to a **bottom-up, stock-level, cross-sectional relative-strength engine**—the Cross-Sectional Quadrant Model—where every stock earns its position individually against the full market, not merely against peers within an arbitrary sector boundary.

3. Previous Approaches: Portfolio Construction & Sizing

3.1 Modern Portfolio Theory (MPT)

1. The Expected Return

The return of a portfolio is the weighted average of the returns of its individual assets. For n assets, the portfolio return $E(R_p)$ is:

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i)$$

2. Risk (Volatility and Correlation)

The risk of a portfolio isn't just the average risk of its stocks; it also depends on covariance. If one goes up when the other goes down, the "zig-zag" cancels out, smoothing your overall journey.

The Efficient Frontier

- **The Goal:** For any given level of risk, you want the highest possible return. Or, for a target return, you want the lowest possible risk.
- **The Curve & Market Portfolio:** Plotting every possible combination of assets forms a region. The upper-left boundary is the Efficient Frontier. Any portfolio sitting on this line is considered "efficient."

> **Why We Pivoted: Covariance Instability.** MPT is famously brittle. Small estimation errors in expected returns $E(R_i)$ or the covariance matrix cause extreme portfolio concentration. It also assumes normal distributions, utterly failing during fat-tailed market crashes.

3.2 Hierarchical Risk Parity (HRP)

HRP builds a portfolio by treating assets as members of a family tree, distributing risk from the "trunk" down to the "leaves" in three main steps:

1. **Hierarchical Clustering:** HRP groups assets by correlation. Similar assets group together first, creating a dendrogram.
2. **Quasi-Diagonalization:** The correlation matrix is rearranged to place similar assets adjacently, forcing the largest correlations along the main diagonal.
3. **Recursive Bisection:** Starting at the top of the tree, assets are split into two clusters and assigned weights based on relative risk (variance). Moving down recursively, highly volatile clusters receive less weight.

> **Why We Pivoted: Return Agnosticism.** While HRP solves MPT's covariance instability by relying strictly on tree-based risk distribution, it *completely ignores expected returns*. It protects the downside but fails to maximize upside alpha, frequently overweighting low-volatility laggards.

3.3 Position Sizing: The Kelly Criterion

While MPT and HRP focus on what to buy, the Kelly Criterion focuses on *how much* to buy to maximize long-term compounding.

The Core Formula
For a bet with a binary outcome, the optimal Kelly fraction (f^*) is calculated as:

$$f^* = \frac{bp - q}{b}$$

Intuition: Size your bet based on your "edge" divided by the "odds."

> **Why We Pivoted: The Estimation Trap.** Kelly is mathematically optimal only if you know your exact probabilities (p). In real markets, overestimating your edge even slightly pushes you into the "over-betting" zone, dramatically increasing the risk of catastrophic drawdown.

> $\rightarrow$ **Solution: The Synthesis.** Relying exclusively on any single analytical framework proved too rigid. We pivoted to a **Genetic Algorithm Portfolio Engine** (Detailed in Step 3). By defining a multi-objective fitness function, the GA evolves position sizes that naturally balance MPT's return-seeking, HRP's risk-clustering, and Kelly's penalty for over-leverage—finding robust global optima without the mathematical fragility of standalone models.

4. Current Strategy — Systematic Alpha Engine

The three-step architecture runs end-of-day on T, executes on the open of $T+1$, and iterates daily across four operational phases.

Phase 1 (End of Day T): Data ingestion, risk metric pre-computation, and market regime detection. **Phase 2 (End of Day T):** Cross-sectional quadrant classification and quadrant-specific stock scoring. **Phase 3 (End of Day T):** Genetic algorithm portfolio optimisation and exit/rebalance evaluation. **Phase 4 (Open of Day T+1):** Efficient execution of the delta between target and current holdings.

Step 1	Step 2	Step 3
Cross-Sectional Quadrant Engine	Quadrant-Specific Stock Scoring	Genetic Algorithm Portfolio Engine

Step 1 — Cross-Sectional Quadrant Engine

Phase 1A: Data Ingestion & Risk Pre-computation

For every stock $i \in \mathcal{U}$ on day T we ingest the OHLCV tuple $(O_i, H_i, L_i, C_i, V_i)$ and GICS sector label s_i. We then pre-compute two risk metrics used across all downstream steps:

14-Day Average True Range.

$$\mathsf{TR}_i(t) = \max\big(H_i - L_i,\ |H_i - C_i^{t-1}|,\ |L_i - C_i^{t-1}|\big)$$

$$\mathsf{ATR}_{14,i}(t) = \frac{1}{14}\sum_{\tau=0}^{13} \mathsf{TR}_i(t-\tau)$$

ATR is used for stop-loss placement and sector-adjusted position sizing.

Rolling Historical Volatility. Compute daily log-returns $r_i(t) = \ln(C_i(t)/C_i(t-1))$.

$$\sigma_i^{\mathsf{hist}}(t) = \sqrt{252}\sqrt{\frac{1}{N}\sum_{\tau=1}^{N}\big(r_i(t-\tau) - \bar{r}_i\big)^2}$$

Used for inverse-volatility weighting and the HRP correlation matrix.

Walk-Forward EMA Signal Pre-computation. For each stock i, a grid search over integer pairs $(f,\, s)$ with $2 \le f < s \le 200$ is run on a rolling 252-day in-sample window. The pair $(\hat{f}_i, \hat{s}_i)$ maximising the in-sample Calmar ratio is selected. The resulting trend signal is lagged by exactly one day to prevent look-ahead bias:

$$\Phi_i(t) = \mathrm{sgn}\big(\mathsf{EMA}_{\hat{f}_i}(C_i,\, t-1)\ -\ \mathsf{EMA}_{\hat{s}_i}(C_i,\, t-1)\big)\ \in\ \{+1, -1\}$$

Any stock with $\Phi_i(t) = -1$ is **hard-rejected** prior to scoring (Step 2).

Phase 1B: Market Regime Detection via Breadth Analysis

Let $\mathcal{U}$ denote the stock universe. For each stock $i \in \mathcal{U}$ define the long-term trend indicator:

$$\delta_i(t) = \begin{cases} 1 & \text{if } P_i(t) > \mathsf{SMA}_{200}(P_i, t) \\ 0 & \text{otherwise} \end{cases}$$

The market breadth at time t is the fraction of stocks in confirmed uptrends:

$$B(t) = \frac{1}{|\mathcal{U}|}\sum_{i \in \mathcal{U}} \delta_i(t), \qquad B(t) \in [0, 1]$$

Smooth breadth with fast ($k = 30$) and slow ($k = 150$) simple moving averages:

$$\mathsf{SMA}_{30}^{B}(t) = \frac{1}{30}\sum_{\tau=0}^{29} B(t-\tau), \qquad \mathsf{SMA}_{150}^{B}(t) = \frac{1}{150}\sum_{\tau=0}^{149} B(t-\tau)$$

Compute the rolling standard deviation of the *spread* $D(t) = \text{SMA}^B_{30}(t) - \text{SMA}^B_{150}(t)$:

$$\text{Regime}(t) = \begin{cases} \textbf{NORMAL} & \text{if } D(t) > +\sigma_D \quad \text{(fast breaks above slow + 1 s.d.)} \\ \textbf{DEFENSIVE} & \text{if } D(t) < -\sigma_D \quad \text{(fast breaks below slow - 1 s.d.)} \\ \text{CARRY} & \text{otherwise (hold prior regime)} \end{cases}$$

where $\sigma_D = \text{std}(D)$ over a trailing 252-day window. The one-standard-deviation band prevents whipsaw regime flips from brief breadth fluctuations.

Interpretation. The breadth signal is a bottom-up aggregation: it asks not "is the index up?" but "what fraction of the investable universe is in a structural uptrend?" This distinction matters because index returns can be driven by a handful of mega-caps while the median stock deteriorates — a market condition that superficially looks bullish but signals fragile internals. The SMA spread $D(t)$ with its $\pm\sigma_D$ band acts as a hysteresis filter, requiring a *sustained, broad-based shift* before a regime flip is declared. This design choice directly addresses the *Equity Correlation Illusion* (RAAM, Section 2): instead of correlations between sector returns (which collapse during crises), we measure regime via cross-sectional breadth, which remains a robust and interpretable signal even at high systemic correlation levels.

Phase 1C: Four-Quadrant Stock Classification

Each stock is placed into one of four quadrants using two binary dimensions evaluated *cross-sectionally* against the full universe each day.

	High Smart-Beta (SB)	**Low Smart-Beta (SB)**
High Market-Cap (MC)	**HH** — Leaders	**HL** — Value/Cyclical
Low Market-Cap (MC)	**LH** — Emerging Stars	**LL** — Laggards

The MC rank of stock i is its percentile rank of market-capitalisation $M_i(t)$ across $\mathcal{U}$, i.e. pct-rank(M_i). The SB rank is its percentile rank of the inverse-volatility-weighted return contribution (from the IVW index of Section 1). A stock is classified "High" if its rank exceeds the cross-sectional median.

In a **NORMAL** regime, the engine targets the **HH, HL, and LH** quadrants (leaders and rising challengers); the LL quadrant is excluded as it captures chronic underperformers. In a **DEFENSIVE** regime, *all* quadrants are eligible but stocks are re-scored on safety metrics (Step 2).

Step 2 — Quadrant-Specific Stock Scoring

Phase 2A: Pre-computation of Alpha Indicators (End of Day T)

Before scoring, the following volume and price-derived alpha features are computed for every stock i. Each indicator is calculated in a single vectorised pass using the full OHLCV history.

ATR$_{14}$	$\text{ATR}_{14}(t) = \frac{1}{14}\sum_{\tau=0}^{13} \max\big(H_{t-\tau} - L_{t-\tau},\, \lvert H_{t-\tau} - C_{t-\tau-1}\rvert,\, \lvert L_{t-\tau} - C_{t-\tau-1}\rvert\big)$
OBV	On-Balance Volume: $\text{OBV}(t) = \text{OBV}(t-1) + V_t \cdot \text{sgn}(C_t - C_{t-1})$
VPT	Volume-Price Trend: $\text{VPT}(t) = \text{VPT}(t-1) + V_t \cdot \frac{C_t - C_{t-1}}{C_{t-1}}$
NVI	Negative Volume Index: $\text{NVI}(t) = \text{NVI}(t-1) \times \big(1 + r_t \cdot \mathbf{1}[V_t < V_{t-1}]\big)$
BBW	Bollinger Bandwidth: $\text{BBW}(t) = \frac{\text{SMA}_{20} + 2\sigma_{20} - (\text{SMA}_{20} - 2\sigma_{20})}{\text{SMA}_{20}} = \frac{4\sigma_{20}}{\text{SMA}_{20}}$

OBV and VPT are used to detect *volume exhaustion* and *OBV divergence* in the multi-factor exit rules (Phase 3C). NVI identifies price movements driven by informed traders (low-volume accumulation). BBW identifies volatility squeezes that precede breakouts, providing timing context for new entries.

Walk-Forward EMA Signal (carried from Step 1). Any stock with $\Phi_i(t) = -1$ is **hard-filtered out** regardless of score.

Phase 2B: NORMAL Regime Scoring — Momentum

We deliberately use a 63-day (3-month) lookback rather than the classical 12-month cross-sectional momentum. The 12-month signal exhibits a well-documented 1-month reversal effect (Jegadeesh & Titman), which contaminates the signal. Our 3-month "Price Inertia" avoids this reversal zone while remaining long enough to capture meaningful trend persistence in Indian mid- and large-cap equities.

$$\text{Price Inertia}_i = \frac{C_i(t) - C_i(t{-}63)}{C_i(t{-}63)} \quad \text{(3-month momentum, no skip-month)}$$

$$\text{Relative Strength}_i = \frac{R_i^{63}}{R_{\mathcal{U},\text{med}}^{63}} \quad \text{(stock return vs. cross-sectional median return)}$$

Both signals are cross-sectionally normalised using the *median absolute deviation* (MAD) rather than standard deviation, making the composite robust to extreme outliers:

$$\tilde{z}(x_i) = \frac{x_i - \text{median}_j(x_j)}{\text{MAD}}, \quad \text{MAD} = \text{median}_j\big|x_j - \text{median}_k(x_k)\big|$$

$$\text{Score}_i^{\text{NORMAL}} = 0.5\ \tilde{z}(\text{Price Inertia}_i) + 0.5\ \tilde{z}(\text{Rel. Strength}_i)$$

Phase 2C: DEFENSIVE Regime Scoring — Safety

In a bear market, capital preservation takes priority. The Gain-to-Pain ratio quantifies the ratio of cumulative gains to cumulative losses over the past year, capturing *asymmetric* return profiles:

$$\text{GtP}_i = \frac{\sum_{t:r_{i,t}>0} r_{i,t}}{\sum_{t:r_{i,t}<0} |r_{i,t}|} \quad \text{(252-day rolling)}$$

$$\sigma_i^{\text{down}} = \sqrt{\frac{1}{T}\sum_{\tau=1}^{T} \min(r_{i,\tau}, 0)^2} \quad \text{(Downside Deviation, analogous to Sortino denominator)}$$

$$\text{Score}_i^{\text{DEFENSIVE}} = 0.5\ \tilde{z}(\text{GtP}_i) - 0.5\ \tilde{z}(\sigma_i^{\text{down}})$$

Stocks that score highest under this metric are those that participated asymmetrically in up-moves while suffering muted losses during down-moves — exactly the defensive quality sought during DEFENSIVE regimes. Note the subtracted downside deviation term: a high σ_i^{down} is penalised, so stocks with historically large downside swings are scored lower even if they have a reasonable GtP ratio.

Phase 2D: Shortlist Construction

Top-ranked stocks (25–100 names) are selected subject to a **20% intra-sector cap**:

$$\sum_{i \in \mathcal{S}_k} \mathbf{1}[\text{selected}] \ \leq\ 0.20 \times N_{\text{target}} \quad \forall \text{ sector } k$$

This prevents any single GICS sector from dominating the shortlist and reintroducing the concentration risk seen in RAAM.

Step 3 — Genetic Algorithm Portfolio Engine

Phase 3A: Fitness Function — Multi-Objective Formulation

Let $\mathbf{w} = (w_1, \ldots, w_n)^\top$ with $\sum_i w_i = 1$, $w_i \geq 0$ be the weight vector. The GA maximises a penalised fitness score:

$$\mathcal{F}(\mathbf{w}) = f_{\text{Kelly}}(\mathbf{w}) \ -\ \lambda_1\, \mathcal{P}_{\text{Sharpe}}(\mathbf{w}) \ -\ \lambda_2\, \mathcal{P}_{\text{HRP}}(\mathbf{w})$$

(i) Continuous Kelly Fraction (geometric growth proxy):

$$f_{\text{Kelly}}(\mathbf{w}) = \mu_p - \tfrac{1}{2}\sigma_p^2, \quad \mu_p = \mathbf{w}^\top \boldsymbol{\mu}, \quad \sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w}$$

where $\boldsymbol{\mu}$ is the vector of expected returns and $\boldsymbol{\Sigma}$ is the sample covariance estimated on a 252-day rolling window.

(ii) Sharpe Ratio Penalty: Let SR* be the Sharpe ratio of the MPT tangency portfolio on the current shortlist.

$$\mathcal{P}_{\text{Sharpe}}(\mathbf{w}) = \max\Big(0,\ 0.6\,\text{SR}^* - \text{SR}(\mathbf{w})\Big)^2$$

If the evolved portfolio's Sharpe falls below 60% of the MPT benchmark, the quadratic penalty sharply increases, nudging the GA back toward efficient solutions.

(iii) HRP Concentration Penalty: Using the HRP dendrogram, partition stocks into two clusters $\mathcal{C}_1, \mathcal{C}_2$ of highly correlated names.

$$\mathcal{P}_{\text{HRP}}(\mathbf{w}) = \max\Big(0,\ \sum_{i \in \mathcal{C}_{\max}} w_i - 0.35\Big)^2$$

where $\mathcal{C}_{\max}$ is the larger cluster. If more than 35% of capital concentrates in a single correlated cluster, the penalty triggers—enforcing diversification without hard weight caps.

Phase 3B: Evolutionary Mechanics

Population	100 randomly initialised weight vectors (Dirichlet-sampled, $\alpha = 1$)
Generations	40 evolutionary cycles
Selection	Tournament selection (size 5); top 20% élites carried forward unchanged
Crossover	Uniform blend: $\mathbf{w}^{\text{child}} = \beta\,\mathbf{w}^A + (1-\beta)\,\mathbf{w}^B$, $\beta \sim U[0,1]$, renormalised
Mutation	Gaussian perturbation $w_i \leftarrow \lvert w_i + \epsilon \rvert$, $\epsilon \sim \mathcal{N}(0, 0.02)$, renormalised
Termination	After 40 generations or when $\Delta\mathcal{F} < 10^{-5}$ for 5 consecutive generations

Convergence & Constraint Feasibility. Because $\sum_i w_i = 1$ is a simplex constraint, all crossover and mutation operations are followed by a renormalisation step $\mathbf{w} \leftarrow \mathbf{w}/\|\mathbf{w}\|_1$ to preserve budget feasibility. The élite carry-over ensures $\mathcal{F}^{(g)} \geq \mathcal{F}^{(g-1)}$ is monotone non-decreasing in the best individual across generations. In practice the GA typically converges in 20–30 generations; the remaining budget provides exploration insurance against premature convergence to local optima in the non-convex fitness landscape induced by the quadratic penalty terms.

The key advantage over a direct convex-optimisation approach (e.g. Sequential Quadratic Programming) is that the GA does *not* require the fitness landscape to be differentiable or convex. The joint Kelly–Sharpe–HRP surface contains flat regions, discontinuities at the penalty activation boundaries (SR$(\mathbf{w}) = 0.6\,$SR* and $\sum_{i \in \mathcal{C}_{\max}} w_i = 0.35$), and multiple local optima; gradient-based methods would stall at such boundaries. The stochastic nature of the GA allows it to explore across these surfaces globally.

Exit Rules. A stock i currently held is flagged for liquidation if *either*:

1. **Hard Stop:** $P_i(t) < P_i^{\text{entry}} - k \cdot \text{ATR}_{14}^{(s_i)}$, where k is a sector-specific multiplier ($k \approx 2.0$–3.5) and s_i denotes stock i's GICS sector.
2. **Multi-Factor Exit:** At least **two** of the following trigger simultaneously: $\Phi_i(t) = -1$ (EMA flip) **AND/OR** OBV divergence detected **AND/OR** VPT trending negative.

Rebalance Triggers. A full portfolio reshuffle is initiated if *any* of the following hold:

- **Calendar:** Start of a new quarter.
- **Regime Switch:** Regime$(t) \neq$ Regime$(t-1)$.
- **Emergency:** Stop-outs reduce the active portfolio below 15 stocks (subject to a 15-day cooldown to prevent overtrading during crash cascades).

Trade Execution Logic. The system computes the delta Δw_i between the target weights w_i^* and current weights w_i^{cur}. Execution priority: SELL $\rightarrow$ TRIM $\rightarrow$ ADD $\rightarrow$ BUY. New positions receive a *fractional Kelly allocation*: only a safe sub-fraction $\kappa < f^*$ of total capital is deployed per position, with the remainder held as a cash buffer.

5. Current Strategy — Phase 4: Execution & Daily Logging

Phase 4 — Efficient Delta Execution (Open of Day T+1)

Trade Delta Computation. Let $\mathbf{w}^* = (w_1^*, \ldots, w_n^*)^\top$ denote the target weights from the GA engine and $\mathbf{w}^{\text{cur}}$ the current portfolio weights (using previous close prices as proxy for the next-day open). The signed delta for each position is:

$$\Delta_i = w_i^* - w_i^{\text{cur}}, \qquad \text{notional change} = \Delta_i \times V_{\text{portfolio}}$$

Trades are sequenced in strict priority order to ensure liquidity is available for new buys:

Priority	Action
1. SELL	Full liquidation of stocks exited by the defence rules (§ Phase 3C).
2. TRIM	Partial reduction for overweight incumbents ($\Delta_i < -\epsilon$).
3. ADD	Increase weight in underweight incumbents ($0 < \Delta_i < \epsilon$).
4. BUY	Initiate new positions using the fractional Kelly allocation κw_i^*.

Fractional Kelly Sizing. Empirically, the full Kelly fraction $f^* = \mu_p/\sigma_p^2$ tends to be too aggressive due to parameter estimation error. We deploy a *sub-Kelly* fraction:

$$f_{\text{safe}} = \kappa \cdot f^*, \quad \kappa \in (0,1), \qquad \text{Cash Buffer} = 1 - \sum_i f_{\text{safe}} \cdot w_i^*$$

The uninvested cash buffer acts as a dry-powder reserve, reducing the depth of drawdowns during regime transitions and allowing rapid re-deployment when breadth recovers.

Execution Benchmark. All simulated trades are executed at the *previous-day closing price* as a proxy for the next-day open. This conservative assumption slightly understates performance but ensures no same-day information leakage into execution prices.

Daily Logging Schema.

Portfolio Equity $V(t)$	Total mark-to-market value of all holdings plus cash.
Drawdown $D(t)$	$D(t) = 1 - V(t)/\max_{\tau \le t} V(\tau)$ — rolling peak-relative drawdown.
Market Breadth $B(t)$	Fraction of universe above 200-day SMA (regime signal).
Regime State	NORMAL / DEFENSIVE on day t for attribution analysis.
Trade Log	Ticker, direction, quantity, price, and delta weight for every executed trade.

6. How to Run the Script (Config + Output Behavior)

How to use it (top of the file)

Edit only this config block:

```
# Period 1 (e.g., 2010-2020)
PARQUET_FILE_P1 = "nse_prices_complete 1.parquet"

# Period 2 (e.g., 2020-2025)
USE_SECOND_DATASET = False
PARQUET_FILE_P2 = "nse_prices_2020_2025.parquet"
```

- Right now keep `USE_SECOND_DATASET = False` (if you don't have 2020–2025 yet).
- Later, when you add the second parquet:
  - Set `USE_SECOND_DATASET = True`

- Set `PARQUET_FILE_P2 = "<your new parquet filename/path>"`

What it will do

- Runs Period-1 fully and writes everything into: `OUTPUT_ROOT/2010_2020/`
- If Period-2 is enabled:
  - Builds using `concat(period1 + period2)` as warm-up history
  - Sets `TRADE_START_DATE = period2_start` so no trades happen before Period-2 begins
  - Reporting/CAGR window is exactly the Period-2 parquet's own date span
  - Writes into `OUTPUT_ROOT/2020_2025/`

7. Strategy Summary & Design Principles

The final architecture synthesises every lesson learned across the six discarded approaches. Each design decision directly corrects a documented failure mode:

Design Choice	Problem Solved	Mechanism
Breadth-based regime	RAAM correlation illusion	Macro-level equity filter before stock selection
MAD Z-scoring	Gaussian fat-tail failures	Median-based normalisation; outlier robust
Walk-forward EMA	In-sample overfitting	Rolling optimisation with strict 1-day lag
Quadrant selection	Intra-sector dispersion	Cross-sectional, not sector-relative ranking
20% sector cap	Concentration risk	Hard constraint enforced before GA inputs
Multi-factor exits	Single-signal false exits	Requires 2+ simultaneous technical failures
GA fitness penalties	MPT fragility + Kelly over-betting	Soft constraints maintain diversification
Sub-Kelly sizing	Catastrophic drawdown	κf^* fraction with explicit cash buffer
15-day rebalance cooldown	Crash cascade over-trading	Emergency reshuffle limited by cooldown timer

Key Hyperparameter Reference

Parameter	Value	Role
SMA$_{200}$	200 days	Long-term trend threshold for breadth calculation
Breadth fast SMA	30 days	Short-term trend of market breadth
Breadth slow SMA	150 days	Long-term trend of market breadth
Regime band	$\pm 1\sigma_D$	Hysteresis threshold for regime flip
ATR lookback	14 days	Stop-loss and volatility proxy
Momentum window	63 days	Price Inertia and Relative Strength
GtP / Downside vol window	252 days	Defensive scoring lookback
GA population size	100	Initial random portfolios per generation
GA generations	40	Maximum evolutionary cycles
Sharpe penalty threshold	60% of MPT SR	Minimum acceptable Sharpe vs. tangency
HRP concentration limit	35%	Max weight in largest correlated cluster
Sector allocation cap	20%	Max fraction of shortlist from any one GICS sector
Rebalance cooldown	15 days	Minimum gap between emergency reshuffles
Min active positions	15	Emergency rebalance trigger threshold
Target shortlist size	25–100	Stocks entering the GA optimisation

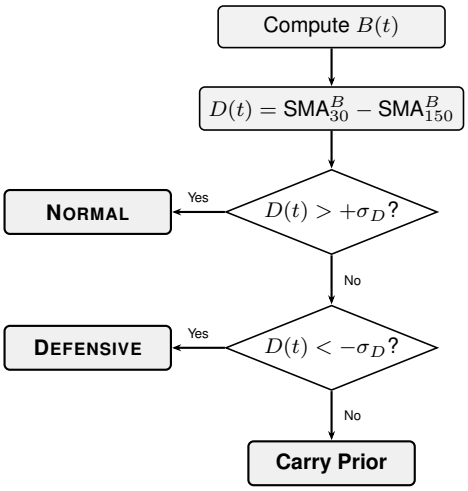
The result is a **fully systematic, regime-aware, long-only equity strategy** that dynamically adapts its stock selection criteria, position sizing, and portfolio construction to the prevailing market environment — without any discretionary overrides or look-ahead information.

Appendix

A.1 Glossary of Mathematical Notation

$\mathcal{U}$	= Investable stock universe
$i,\, j$	= Stock indices; $i \in \mathcal{U}$
$t,\, \tau$	= Time indices (daily)
T	= EOD compute date; trades on $T{+}1$ open
$C_i,\, H_i,\, L_i,\, V_i$	= Close, High, Low, Volume
$r_i(t)$	= Log-return $\ln\!\big(C_i(t)/C_i(t{-}1)\big)$
$M_i(t)$	= Market capitalisation
σ_i^{ann}	= Annualised historical volatility
$\text{ATR}_{14,i}$	= 14-day Average True Range
$B(t)$	= Market breadth (fraction above 200-SMA)
$D(t)$	= Breadth spread $\text{SMA}_{30}^{B} - \text{SMA}_{150}^{B}$
σ_D	= Std. of $D(t)$ over 252-day window
$\Phi_i(t)$	= EMA trend signal $\in \{+1, -1\}$, 1-day lag
$\tilde{z}(\cdot)$	= MAD cross-sectional Z-score
GtP_i	= Gain-to-Pain ratio (252-day window)
σ_i^{down}	= Downside deviation
$\mathbf{w},\, w_i$	= Weight vector; weight of stock i
$\boldsymbol{\mu},\, \boldsymbol{\Sigma}$	= Expected returns; covariance matrix
$\mathcal{F}(\mathbf{w})$	= GA multi-objective fitness function
$f^*,\, f_{\text{safe}}$	= Full and fractional Kelly fraction
SR^*	= MPT tangency Sharpe (benchmark)
$\mathcal{C}_1, \mathcal{C}_2$	= HRP dendrogram clusters
λ_1, λ_2	= GA penalty weights
κ	= Sub-Kelly scaling factor $\in (0, 1)$

A.2 Regime Transition Flowchart



A.3 GA Penalty Activation

The two penalty terms in $\mathcal{F}(\mathbf{w})$ are *quadratic* beyond their thresholds — a soft wall that nudges the GA rather than a hard constraint.

Penalty	Activation & Effect
$\mathcal{P}_{\text{Sharpe}}$	Fires when $\text{SR}(\mathbf{w}) < 0.6\,\text{SR}^*$. Pushes offspring back toward the MPT efficient frontier, preventing risk-free dominated solutions.
$\mathcal{P}_{\text{HRP}}$	Fires when $\sum_{i \in \mathcal{C}_{\max}} w_i > 0.35$. Redistributes weight away from the dominant correlated cluster, enforcing HRP-style diversification.

Both penalties vanish when constraints are satisfied, so they add zero cost to a well-diversified solution — the GA is only penalised when it strays into undesirable regions.

References

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